

Multiple Choice (10 marks)

- Which of the following is lower for argon than for neon?
 - Melting point
 - Boiling point
 - Polarizability
 - First ionization energy
- Calculate the equilibrium polarization of ^{13}C nuclei in a 20.0 T magnetic field at 300 K.
 - 10.8×10^{-5}
 - 4.11×10^{-5}
 - 3.43×10^{-5}
 - 1.71×10^{-5}
- Calculate the Q-factor for an X-band EPR cavity with a resonator bandwidth of 1.58 MHz.
 - $Q = 1012$
 - $Q = 2012$
 - $Q = 3012$
 - $Q = 6012$
- The number of allowed energy levels for the ^{55}Mn nuclide are:
 - 3
 - 5
 - 8
 - 4
- When the following equation is balanced, which of the following is true? $\text{MnO}_4^- + \text{I}^- + \text{H}^+ \rightleftharpoons \text{Mn}_2^+ + \text{IO}_3^- + \text{H}_2\text{O}$
 - The $\text{I}^- : \text{IO}_3^-$ ratio is 3:1.
 - The $\text{MnO}_4^- : \text{I}^-$ ratio is 6:5.
 - The $\text{MnO}_4^- : \text{Mn}_2^+$ ratio is 3:1.
 - The $\text{H}^+ : \text{I}^-$ ratio is 2:1.

True / False (10 marks)

Answer True or False

6. True or False: K^+ has a smaller ionic radius than Sc^{3+} because it is a smaller atom in the periodic table.
7. True or False: Infrared spectroscopy is a light-scattering technique that primarily analyzes molecular vibrations.
8. True or False: NH_3 is considered the strongest base in liquid ammonia because it has a high tendency to donate protons.
9. True or False: Silicon is most commonly found in nature as an oxide, primarily in the form of silica and silicate minerals.
10. True or False: Doubling the volume of a buffer solution by adding water significantly increases its pH.

Long Form Response (20 marks)

Answer each question with a clear and complete explanation.

11. Discuss the factors that contribute to the differing relaxation times, T_1 and T_2 , of a nucleus, with an emphasis on the influence of molecular motions at the Larmor frequency.
12. Discuss the concept of ionic strength and analyze why a 0.050 M solution of $AlCl_3$ exhibits higher ionic strength compared to other ionic solutions.

End of Paper